Tennis Bio-Mechanics : the principle of "opposite force"

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This article presents the biomechanics principle of "opposite force" and its application in tennis.

This principle refers to the Newton's third law of motion: for every action there is an equal and opposite reaction. This means that for every force there is a reaction force that is equal in size, but opposite in direction. That is to say that whenever an object pushes another object it gets pushed back in the opposite direction equally hard.

Tennis players use the ground reaction forces both to move on the court and to initiate all tennis shots. They push the ground in different ways (direction and intensity) depending on the type of movement or shot they have to perform. Understanding how the players interact with the ground is important for coaches in order to plan both efficient technical and physical training.

Split step

The first goal of the split step is to put the body in motion in a specific way that enables the best time push in the direction required by the incoming shot. In a perfectly timed split the player leaves the ground just slightly before his opponent makes contact in order to take information about the incoming ball while being in the air. It allows to land first on the outside foot (the foot furthest from the direction of movement) and start with a quick and hard push off. **Figure 1 shows how the outside foot drives the body weight until it is positioned over the inside foot during lateral movement.**

Figure 1

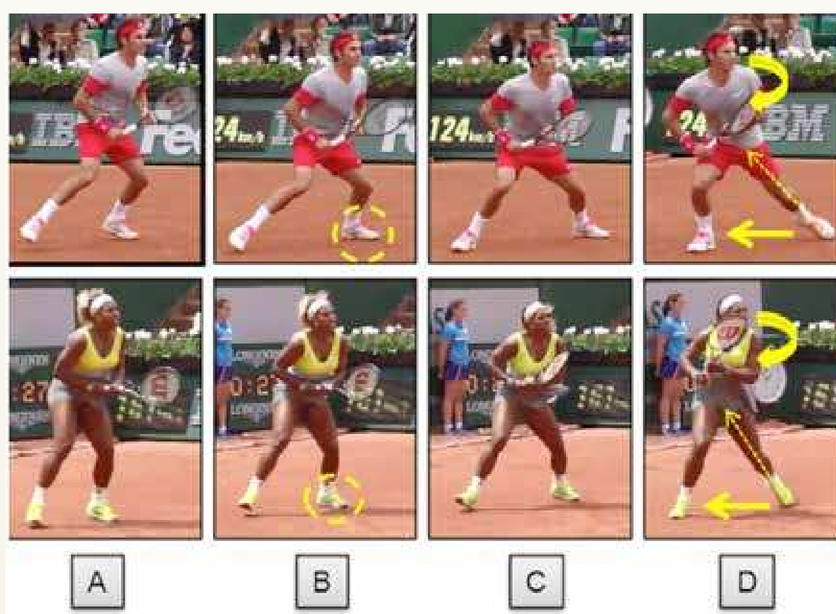
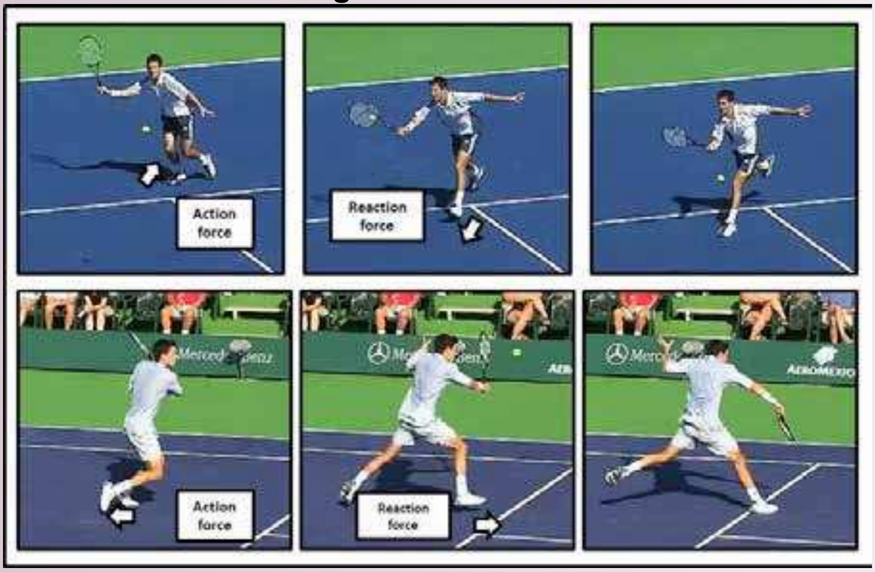


Figure 2

Recovery

Figure 2 shows how the player uses the ground to recover laterally from a wide position. He performs a cross over step involving different hip joint actions.

Figure 3



Shot production

All tennis shots start from the ground. The players generate forces by pushing the ground and they are transferred up through the trunk to the hitting arm. Both horizontal and vertical force components are present but their respective share depends on the type of shots.

Figure 3 shows how the player pushes the ground with the back leg mainly horizontally to perform a volley.

Figure 4

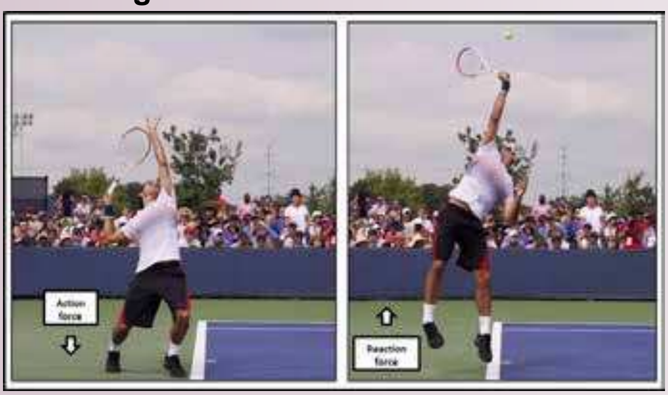


Figure 4 shows how the player pushes the ground mainly vertically to perform a serve.

Figure 5

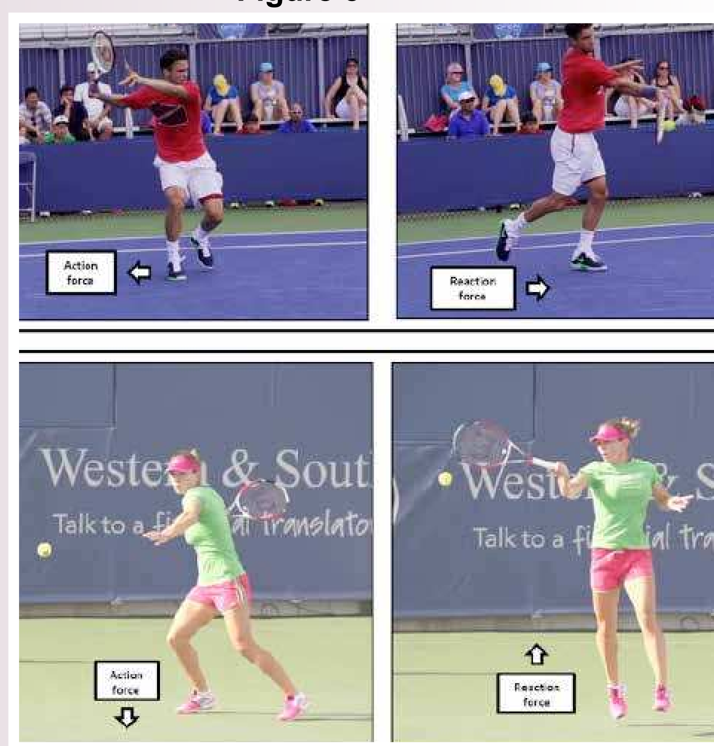


Figure 5 shows how the players push the ground with the back leg mainly horizontally to perform an aggressive forehand shot using a neutral stance and mainly vertically when using an open stance.

Figure 6

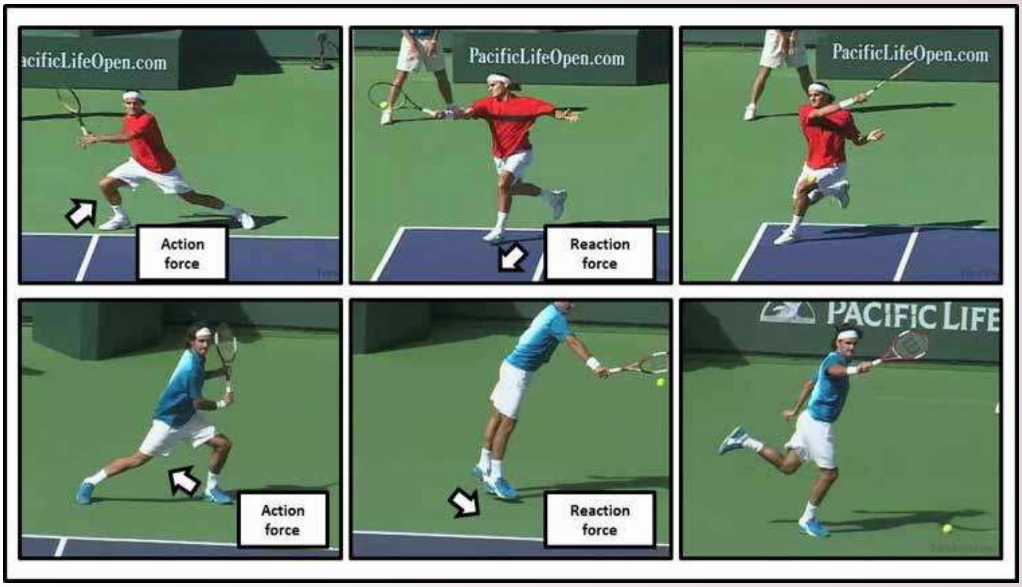


Figure 6 shows how the player pushes the ground both horizontally and vertically to perform a return at shoulder height level.

Conclusion:

The "opposite force" principle is present in all tennis movements and shots production and its understanding should allow coaches to focus on exercices aiming at improve the ability of the player to better use the ground. The exercises can be oriented more on the technical or physical component depending on the player's needs. The next article will address the biomechanics principle of balance and how it's used by players to move in the tennis court and hit shots more efficiently.

WTA Future Star

The WTA Future Stars features competition for young girls across the Asia-Pacific region with a vision to promote high-level athlete development, health, fitness and empowerment of women through tennis. The qualifiers from each country will compete at the Kallang Tennis Centre on October 17-19 in conjunction with the BNP Paribas WTA Finals Singapore presented by SC Global, with the finals being played on Center Court at the Singapore Indoor Stadium on October 20.

Pune girl Salsa Aher represents our country at WTA Future Stars Tournament to be held in Singapore from October 16 to 23. The tournament would see top Under-16 and Under-14 players from 24 countries (one from each country) taking part. The tournament has been divided into four groups of six players each in Round Robin format and top player from each group would qualify for the semi-finals.

Salsa idolises ace tennis star Garbine Muguruza. "I follow her game style a lot. In fact, my game is a lot like her's – the way she hits the ball, her explosive split step, her footwork, service action," said Salsa. Salsa practices at Bounce Academy and is sponsored by Lakshya/GIC and MSLTA Vision program. A year ago, her ITF Jr world ranking was around 1,000 and today she is ranked 315